

DOMPurify: Prototype Pollution to XSS Bypass via CUSTOM_ELEMENT_HANDLING Fallback

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Content

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DOMPurify: Prototype Pollution to XSS Bypass via CUSTOM_ELEMENT_HANDLING Fallback

- Advisory: GHSA-v9jr-rg53-9pgp
- CVE: CVE-2026-41238
- Severity: medium
- Published: 2026-04-22
- Updated: 2026-04-27

Affected

- `dompurify` (npm): `>= 3.0.1, < 3.4.0`, fixed in 3.4.0

Description

Summary

DOMPurify versions 3.0.1 through 3.3.3 (latest) are vulnerable to a prototype pollution-based XSS bypass. When an application uses `DOMPurify.sanitize()` with the default configuration (no `CUSTOM_ELEMENT_HANDLING` option), a prior prototype pollution gadget can inject permissive `tagNameCheck` and `attributeNameCheck` regex values into `Object.prototype`, causing DOMPurify to

allow arbitrary custom elements with arbitrary attributes — including event handlers — through sanitization.

Affected Versions

- **3.0.1 through 3.3.3** (current latest) — all affected
- **3.0.0 and all 2.x versions** — NOT affected (used `Object.create(null)` for initialization, no `|| {}` reassignment)
- The vulnerable `|| {}` reassignment was introduced in the 3.0.0 3.0.1 refactor
- This is **distinct** from GHSA-cj63-jhhr-wcxv (USE_PROFILES Array.prototype pollution, fixed in 3.3.2)
- This is **distinct** from CVE-2024-45801 / GHSA-mmhx-hmjr-r674 (`__depth` prototype pollution, fixed in 3.1.3)

Root Cause

In `purify.js` at line 590, during config parsing:

```
CUSTOM_ELEMENT_HANDLING = cfg.CUSTOM_ELEMENT_HANDLING || {};
```

When no `CUSTOM_ELEMENT_HANDLING` is specified in the config (the default usage pattern), `cfg.CUSTOM_ELEMENT_HANDLING` is `undefined`, and the fallback `{}` is used. This plain object inherits from `Object.prototype`.

Lines 591-598 then check `cfg.CUSTOM_ELEMENT_HANDLING` (the original config property) — which is `undefined` — so the conditional blocks that would set `tagNameCheck` and `attributeNameCheck` from the config are never entered.

As a result, `CUSTOM_ELEMENT_HANDLING.tagNameCheck` and `CUSTOM_ELEMENT_HANDLING.attributeNameCheck` resolve via the prototype chain. If an attacker has polluted `Object.prototype.tagNameCheck` and `Object.prototype.attributeNameCheck` with permissive values (e.g., `/.*/`), these polluted values flow into DOMPurify's custom element validation at lines 973-977 and attribute validation, causing all custom elements and all attributes to be allowed.

Impact

- **Attack type:** XSS bypass via prototype pollution chain
- **Prerequisites:** Attacker must have a prototype pollution primitive in the same execution context (e.g., vulnerable version of `lodash`, `jQuery.extend`, `query-string` parser, `deep merge` utility, or any other PP gadget)
- **Config required:** Default. No special DOMPurify configuration needed. The standard `DOMPurify.sanitize(userInput)` call is affected.
- **Payload:** Any HTML custom element (name containing a hyphen) with event handler attributes survives sanitization

Proof of Concept

```
// Step 1: Attacker exploits a prototype pollution gadget elsewhere in the application
Object.prototype.tagNameCheck = /.*/;
Object.prototype.attributeNameCheck = /.*/;

// Step 2: Application sanitizes user input with DEFAULT config
const clean = DOMPurify.sanitize('<x-x onfocus=alert(document.cookie) tabindex=0 autofocus>');

// Step 3: "Sanitized" output still contains the event handler
console.log(clean);
// Output: <x-x onfocus="alert(document.cookie)" tabindex="0" autofocus="">

// Step 4: When injected into DOM, XSS executes
document.body.innerHTML = clean; // alert() fires
```

Tested configurations that are vulnerable:

Call Pattern	Vulnerable?
<code>DOMPurify.sanitize(input)</code>	YES
<code>DOMPurify.sanitize(input, {})</code>	YES
<code>DOMPurify.sanitize(input, { CUSTOM_ELEMENT_HANDLING: null })</code>	YES
<code>DOMPurify.sanitize(input, { CUSTOM_ELEMENT_HANDLING: {} })</code>	NO (explicit object triggers L591 path)

Suggested Fix

Change line 590 from:

```
CUSTOM_ELEMENT_HANDLING = cfg.CUSTOM_ELEMENT_HANDLING || {};
```

To:

```
CUSTOM_ELEMENT_HANDLING = cfg.CUSTOM_ELEMENT_HANDLING || create(null);
```

The `create(null)` function (already used elsewhere in DOMPurify, e.g., in `clone()`) creates an object with no prototype, preventing prototype chain inheritance.

Alternative application-level mitigation:

Applications can protect themselves by always providing an explicit `CUSTOM_ELEMENT_HANDLING` in their config:

```
DOMPurify.sanitize(input, {  
  CUSTOM_ELEMENT_HANDLING: {  
    tagNameCheck: null,  
    attributeNameCheck: null  
  }  
});
```

Timeline

- **2026-04-04:** Vulnerability discovered during automated DOMPurify fuzzing research (Fermat project)
- **2026-04-04:** Confirmed in Chrome browser with DOMPurify 3.3.3
- **2026-04-04:** Verified distinct from GHSA-cj63-jhhr-wcxv and CVE-2024-45801
- **2026-04-04:** Advisory drafted, responsible disclosure initiated

Credit

<https://github.com/trace37labs>

References

- <<https://github.com/cure53/DOMPurify/security/advisories/GHSA-v9jr-rg53-9pgp>>
- <<https://github.com/cure53/DOMPurify/releases/tag/3.4.0>>
- <<https://nvd.nist.gov/vuln/detail/CVE-2026-41238>>
- <<https://github.com/advisories/GHSA-v9jr-rg53-9pgp>>

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